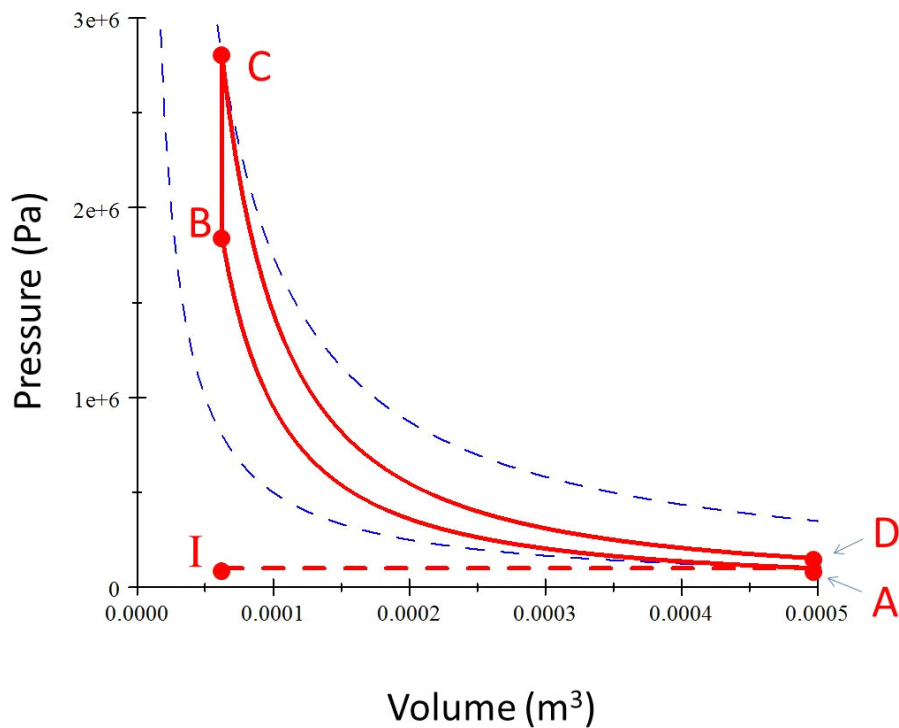
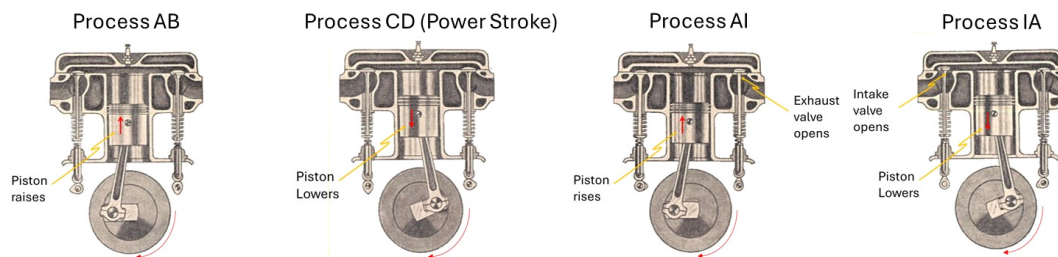


1 An example: The Otto Cycle

A cyclic process is made from more than one thermodynamic process. The Otto cycle technically has six processes. To design an engine we need to describe all of these processes and make sure they work together to produce a net useful work.



The PV diagram above shows the Otto cycle. This cycle is approximately what happens in a gasoline engine, like those in cars or, more like what we find in lawn mowers. It consists of four main processes and two pseudo processes. All of these processes correspond to a piston moving in a cylinder (our favorite thermodynamic system!). Here is a picture of what is happening in the engine.



To understand the Otto cycle let's do the job of finding ΔE_{int} , Q , and w_{int} for each part of the cycle and then describe the state of the gas at the beginning and end of each cycle by finding P , V , n , T , and E_{int} . For our engine, suppose the compression ratio is

$$\frac{V_A}{V_B} = 8.00$$

Let's take

$$\begin{aligned} V_A &= 500 \text{ cm}^3 = 0.0005 \text{ m}^3 \\ P_A &= 100 \text{ kPa} \\ T_A &= 293 \text{ K} \end{aligned}$$

Then we can find V_B

$$V_B = \frac{V_A}{8} = 62.5 \text{ cm}^3 = 6.25 \times 10^{-5} \text{ m}^3$$

and we will need to measure the temperature at C . Suppose we measure

$$T_C = 1023 \text{ K}$$

and the intake temperature to be

$$T_I = 293 \text{ K}$$

Further suppose our engine uses a diatomic ideal gas with

$$\begin{aligned} C_V &= \frac{5}{2}R \\ R &= 8.314 \frac{\text{J}}{\text{mol K}} \end{aligned}$$

so

$$\gamma = 1.4$$

Let's follow the process through to see what the state variables are at each point on the PV diagram, and what Q_h and Q_c are for each process, and what the work, W , is and what the change in internal energy will be, ΔE_{int} . Finally, we will find the efficiency of the engine.